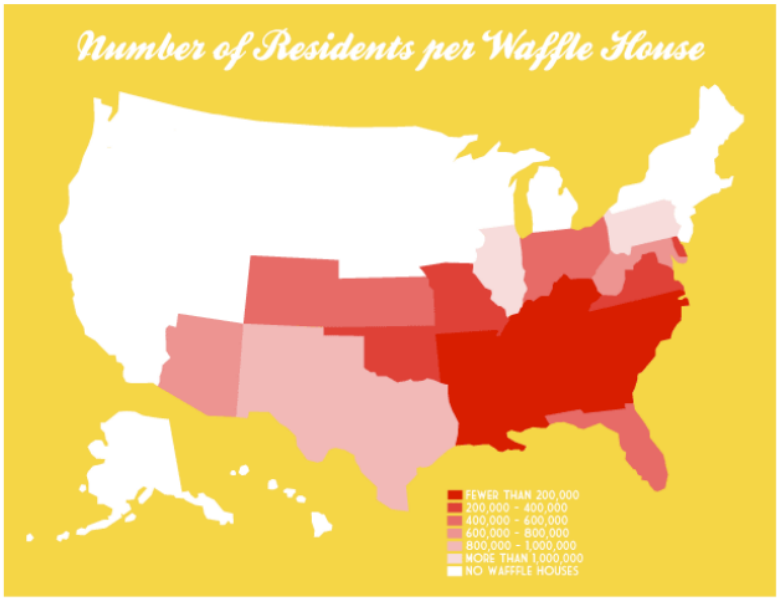
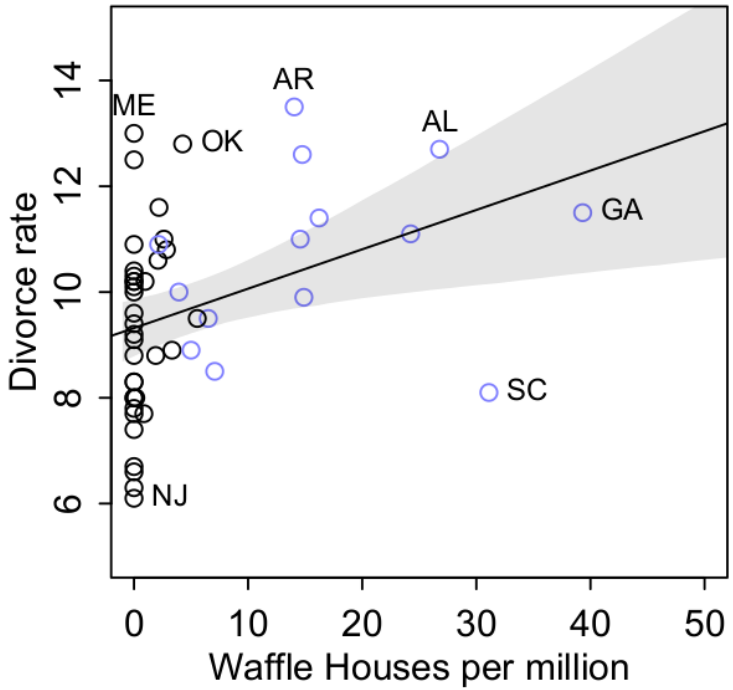
Lecture 13: Multiple regression models, spurious associations

- Multiple regression models: go beyond one predictor; we very quickly and discretely did one continuous plus one factor last week.
- Good aspects:
  - Reveal spurious associations.
  - Uncover masked associations.
- Bad aspects:
  - Cause spurious associations.
  - Hide real associations :-/.
- To do our best and try to stick to the good side, we will learn some basics of causal inference this week.
  - DAGs (Directed Acyclic Graphs).
  - Forks, pipes, colliders and some more plumbing.
  - The backdoor criterion.

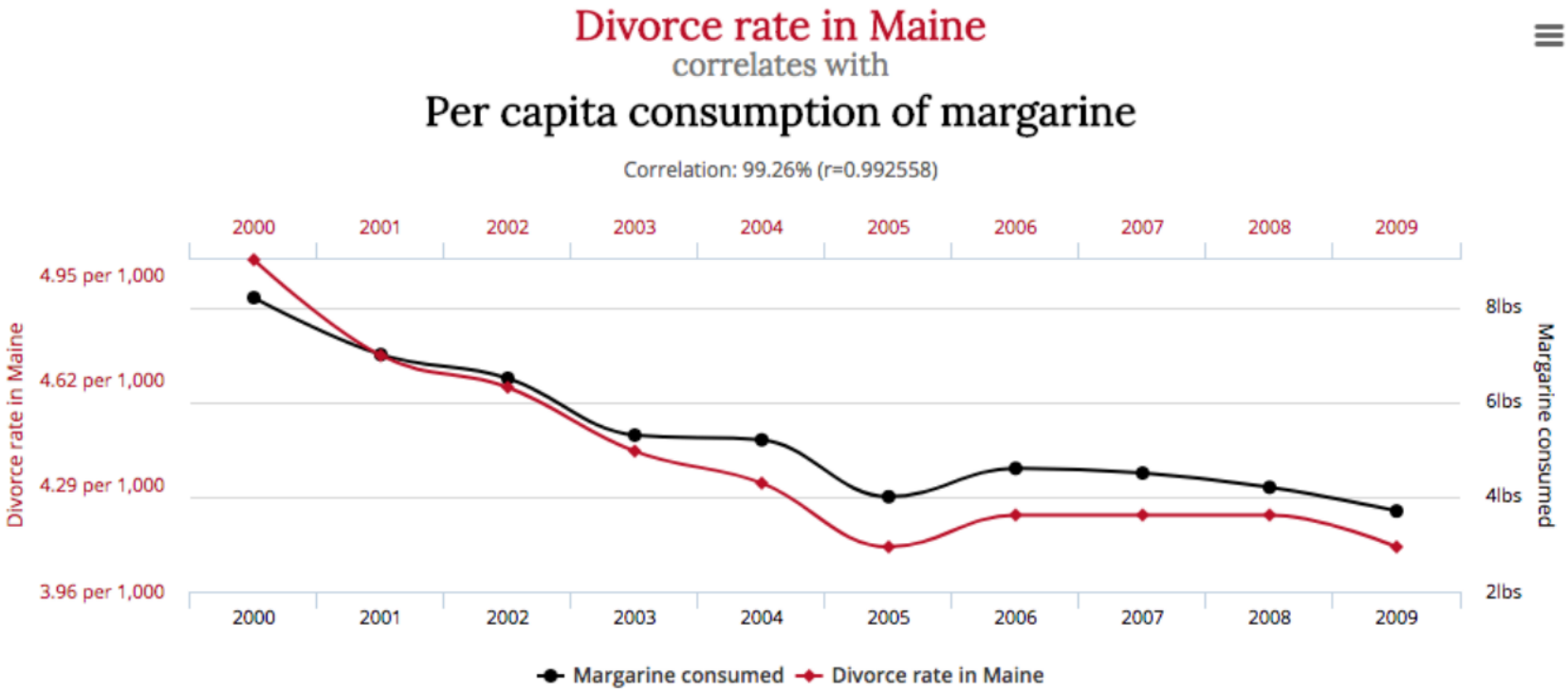
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Does Waffle Houses cause divorce??





Correlation is commonplace



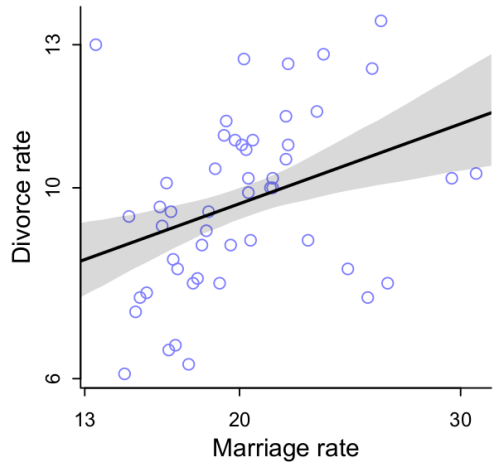
And if you want more to impress your friends at a BBQ, the source is: http://www.tylervigen.com/spurious-correlations

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Spurious associations

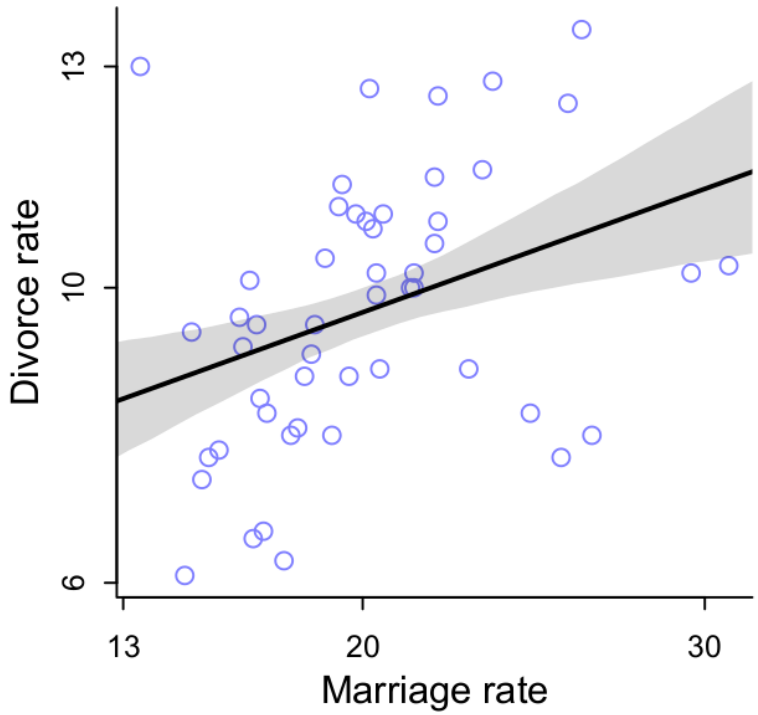
- You've heard that before: correlation does not imply causation.
- Personal note 1: but it doesn't discard it either.
- Hope you are seated: causation does not imply correlation.
- Causation implies conditional correlation (up to linearity issues).
- We need more than just statistical models to answer causal questions.

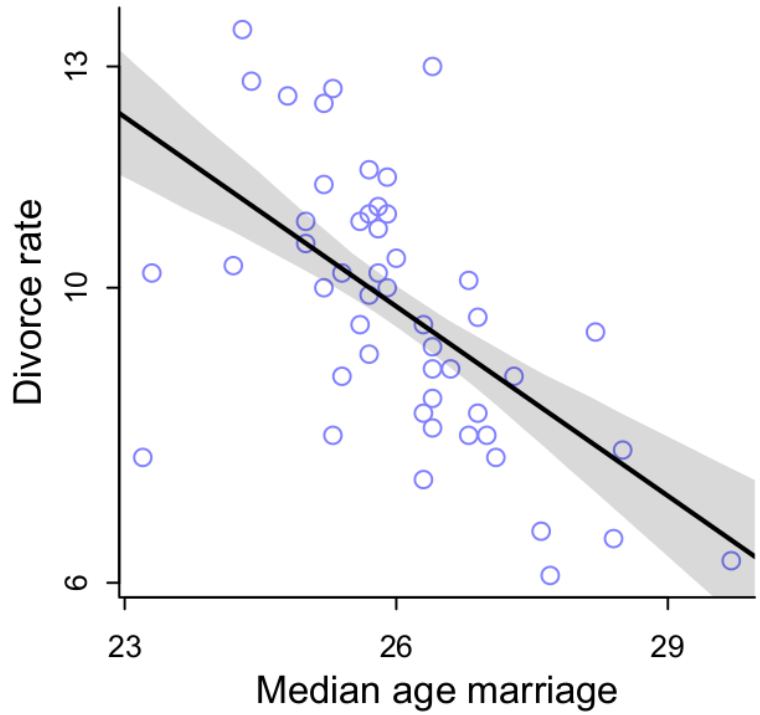
- Question: Does marriage cause divorce? Or vice versa??



Spurious associations

Trying to shed some light





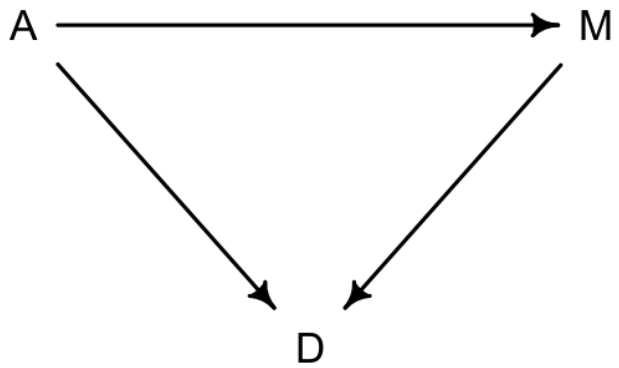
Resolving a spurious association

There are multiple causes of divorce

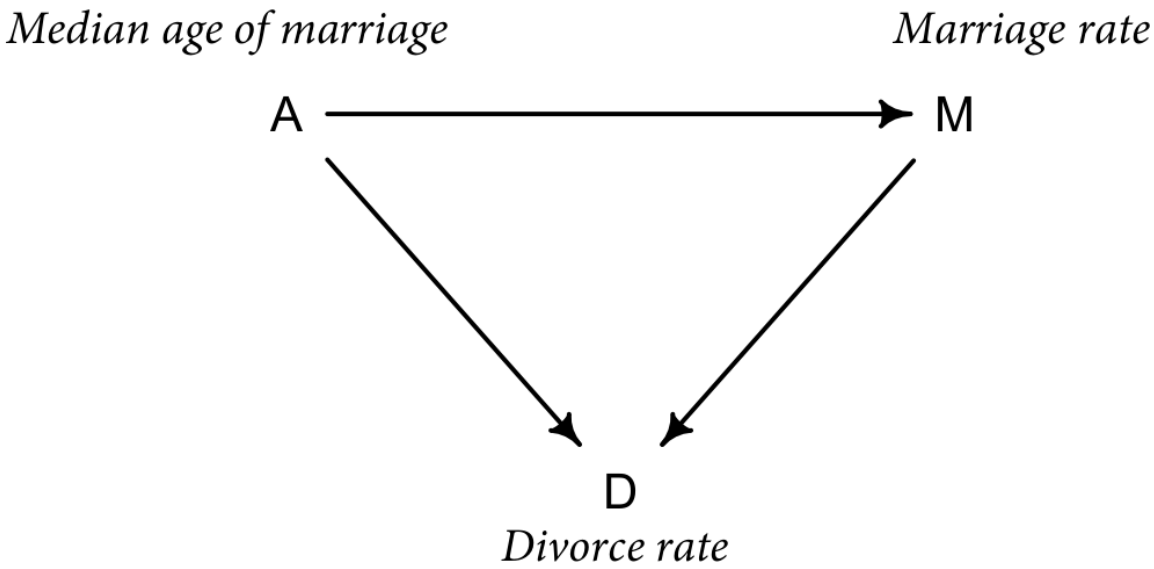
- We want to know:
  - What is the value of a predictor, once we know the other predictors?
- In out case: what is the value of knowing the marriage rate in a state, once we already know the median age at marriage?
- Or: what is the value of knowing the median age at marriage, once we know the marriage rate?
- A DAG can help

DAGs will be your friends

- Directed Acyclic Graphs are good tools for causal models
  - Directed: arrows so $A \rightarrow B$ for A causes B.
  - Acyclic: no loops (a limitation in some field; can resort to Dynamic DAGs, unrolling the time, but needs lots of data)
  - Graphs, we should have started by this, are made of nodes (variables) and (directed) edges for causal relationships.
  - We may encounter partially directed graphs.
- Unlike statistical models, DAGs have causal implications



Deciphering the elements of the DAG

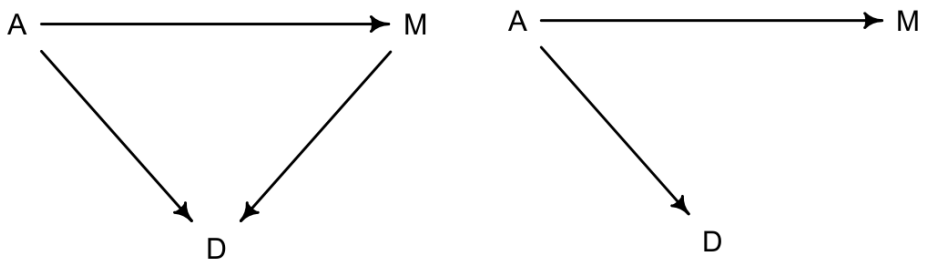


Implications:

1. M is a function of A,
2. D is a function of A and M,
3. (A is not a function of M and/or D)
4. The total causal effect of A on D has two paths:
   1. $A \rightarrow M \rightarrow D$ and
   2. $A \rightarrow D$

Best DAG?

- Give an association $M \longleftrightarrow D$, one cannot tell immediately the difference between:



- The conditional association $M \longleftrightarrow D|A$ is needed.
- How do we do this with multivariate regression?

Disentangling associations with multivariate regression

$$
\begin{aligned}
D_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_M M_i + \beta_A A_i
\end{aligned}
$$

- Standardise all variables: D, M and A
- We expect α to be near 0:

$$\alpha \sim \text{Normal}(0, 0.2)$$

- Slopes should not produce impossibly strong relationships:

$$
\begin{aligned}
\beta_M &\sim \text{Normal}(0, 0.5) \\
\beta_A &\sim \text{Normal}(0, 0.5)
\end{aligned}
$$

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Disentangling associations with multivariate regression

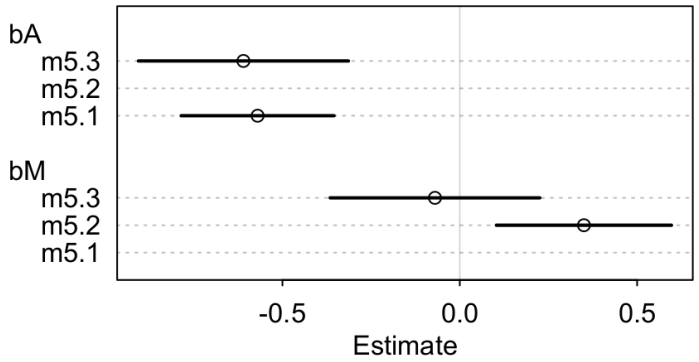
Implementation

```
m5.3 <- quap(list(
  D ~ dnorm(mu, sigma),
  mu <- a+bM*M+bA*A,
  a ~ dnorm(0,0.2),
  bM ~ dnorm(0,0.5),
  bA ~ dnorm(0,0.5),
  sigma ~ dexp(1)
), data = d)
precis(m5.3)
       mean   sd  5.5% 94.5%
a      0.00 0.10 -0.16  0.16
bM    -0.07 0.15 -0.31  0.18
bA    -0.61 0.15 -0.85 -0.37
sigma 0.79 0.08  0.66  0.91
```

(`m5.1` and/or `m5.2` are also fitted)

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Comparison of regression coefficient estimates

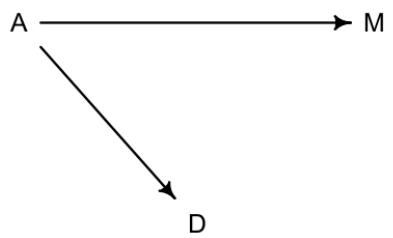


`m5.1`: $D \sim A$ regressing on Age of marriage only.
`m5.2`: $D \sim M$ regressing on marriage rate only.
`m5.3`: $D \sim A + M$ multivariate (or multiple) regression.

Comparison of regression coefficient estimates

Conclusions

- Once we know the median age at marriage, there is little additional value in knowing marriage rate.
- Once we know marriage, there is still value in knowing the median age at marriage.
- If we don't know the median age at marriage, there is value in knowing the marriage rate. *M* is predictive, not causal.
- Hence the plausible model is (if we assume no other causal variables missing):

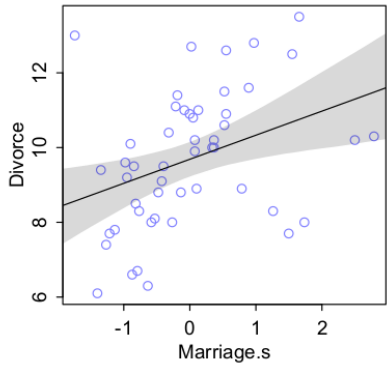


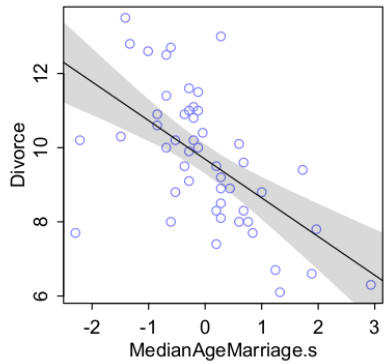
- Let's validate.

Validation

Posterior predictions

- I will assume you have done the prior predictions and are happy with the priors (do it, refine them if needed, feel free to discuss it on Stream!)
- Lots of plotting options:
  1. Predictor residual plots,
  2. Counterfactual plots and
  3. Posterior prediction plots.

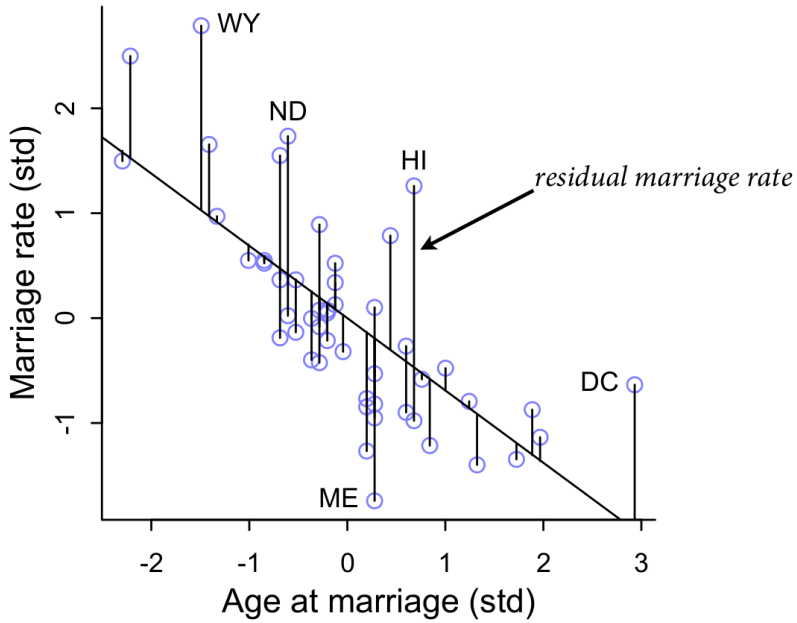




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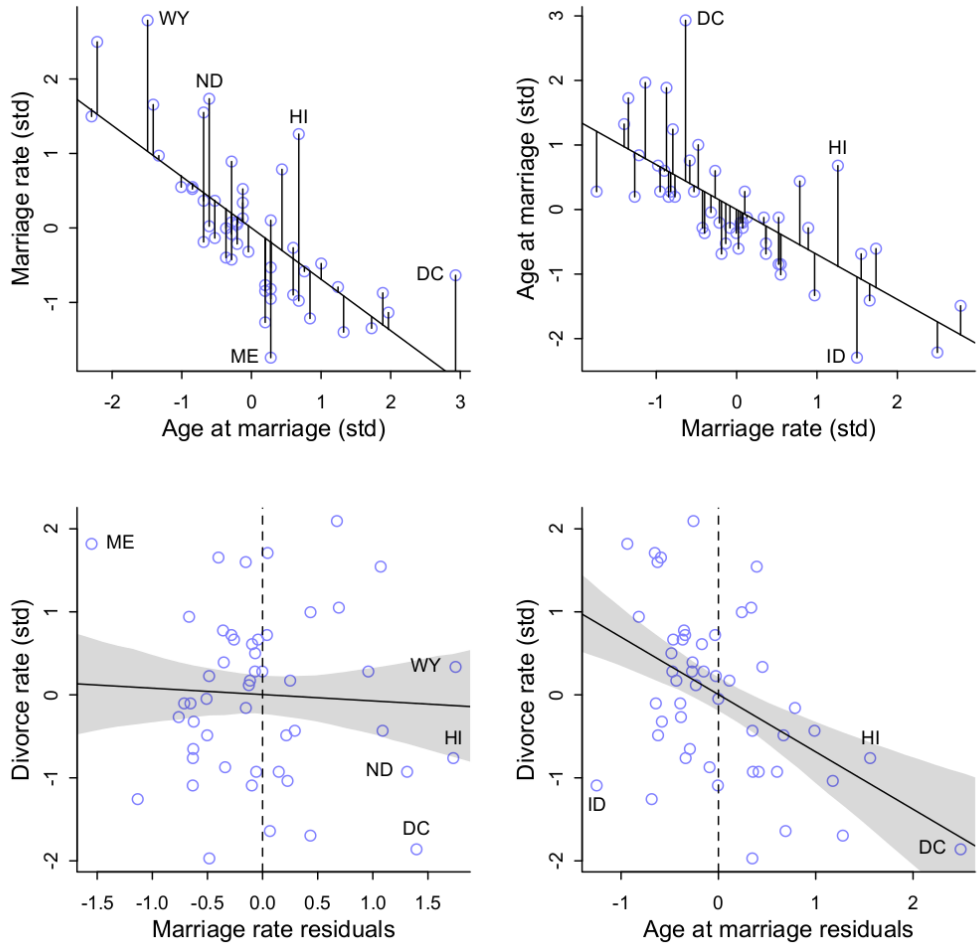
Predictor residual plots

- Goal: show the association of each predictor with the outcome "controlling" for the other predictors.
- Useful intuition.
- Never analyse residuals.
- Recipe:
  1. Regress predictor on other predictors.
  2. Compute predictor residuals.
  3. Regress outcome on residuals.



Predictor residual plot

Altogether:



(Fig. 5.4 in SR book)

Explanations

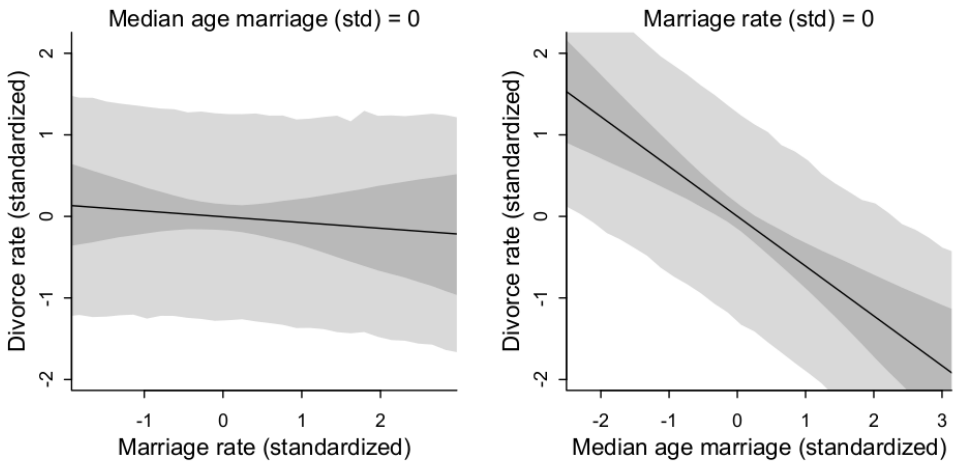
Statistical "control" in multiple regression

- Residuals are leftover variations in one variable after taking out the relationship with other predictors (and are parameters, not data, see box in SR book)
- Regression models measure the remaining association with each predictor with the outcome *after already knowing the other predictors*.
- No magic, this is how they are built.
- Don’t over-conclude. Some states may exhibit a relationship between marriage and divorce rates.
- We cannot make strong causal inference conclusions from averaged trends, we need data on individuals. So let’s look at...

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Counterfactual plots

- Their goal is to explore model implications for the outcome when (i) fixing the values of all predictors but one and (ii) varying the values of the remaining predictor.
- Then you can vary the fixed predictor values. This can be time- and organisation-consuming if there are many predictors!

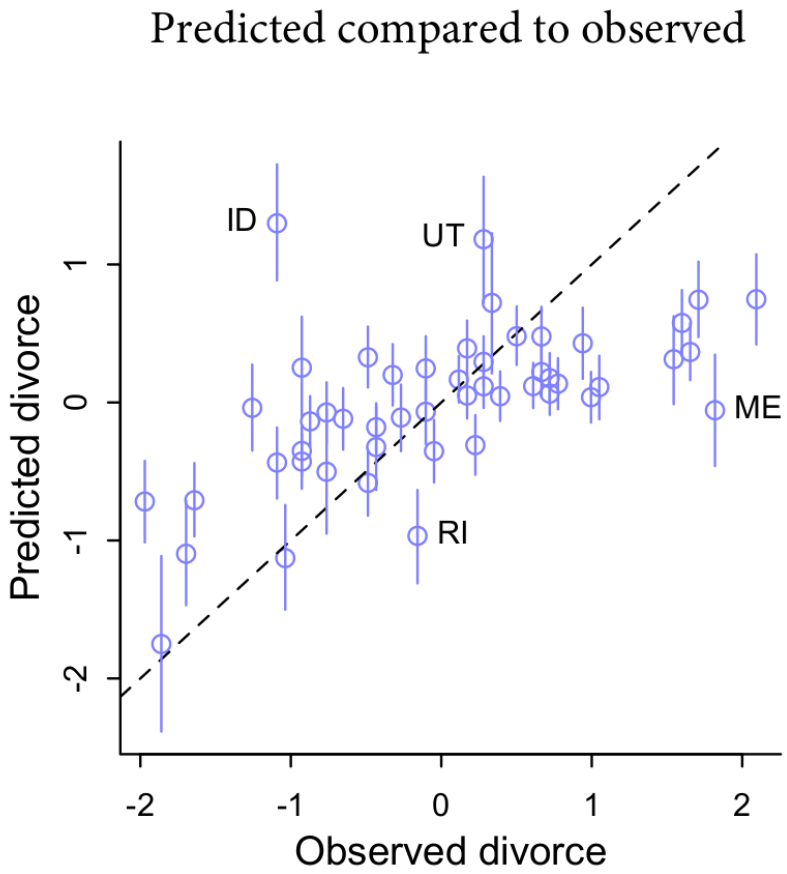


(Fig. 5.5 in SR book)

Posterior prediction checks

- Goal: compute implied predictions for *observed* cases.
  - Check model fit; golems do make mistakes (coding, approximations not valid...)
  - Find model failure to stimulate new ideas.
- Always average over the posterior distribution.
  - Using the posterior mean only leads to overconfidence and neglecting the uncertainty around parameter estimation.
  - Instead, embrace the uncertainty! This is what the posterior predictive predictions do.

Posterior predictive checks



(Fig. 5.6 in SR book)

Conclusive remarks

- We have seen an example, where multiple regression was disentangling the association between two predictors and an outcome.
- Tomorrow, we are going to see an example of a masked association.